National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Electrical Network Analysis	Course Code:	EE-2004
Program:	BS (EE)	Semester:	Fall 2023
Duration:	60 Minutes	Total Marks:	40
Paper Date:	11/11/2023	Weight	15%
Section:	All	Page(s):	1
Exam Type:	Midterm -2	CLO	2 & 3

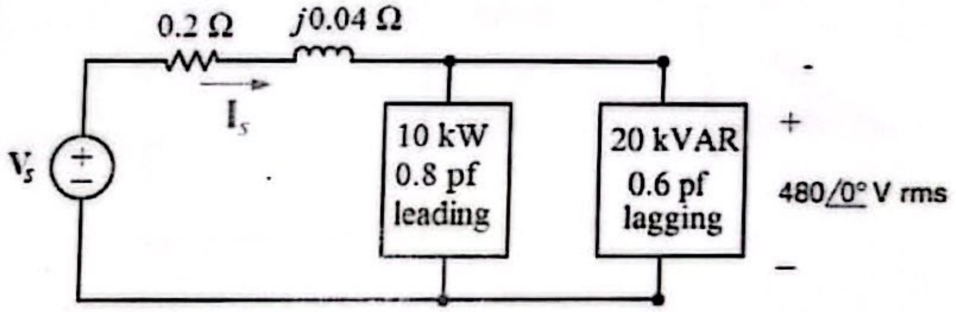
Student : Name: [redacted] **Section:**

Instruction/Notes:

- *Attempt all the questions and solve all parts of a question together*
- *Carefully understand the question before attempting*
- *Show all calculations, direct answers are not acceptable.*

Question # 1 **CLO 2** **[8 + 4 + 4 + 4 = 20 marks]**

The two loads in the circuit shown below can be described as follows:
Load 1 absorbs an average power of 10.0 kW at a leading power factor of 0.8 and Load 2 absorbs 20.0 kVAR at a lagging power factor of 0.6. A voltage of $480\angle 0^\circ$ V_{rms} appears across the loads as shown.



a) Construct the power triangle of each load and determine the power factor of the loads in parallel. Also calculate the impedance of the combined load.
b) Compute the magnitude of the current, I_S and theaverage power loss in the transmission line.
c) Calculate the value of source voltage, V_S and the power factor of the source
d) Given that the frequency of the source is 60 Hz, compute the value of the capacitor that would correct the power factor to 1.0 if placed in parallel with the loads.

Question # 2 **CLO 3** **[5 + 5 + 5 + 5 = 20 marks]**

The phase voltage at the terminals of the a-phase of a balanced three-phase Y-connected load is $1600\angle 0^\circ$ V_{rms}. The load has an impedance of $12 + j9\ \Omega/\varphi$ and is fed from a line having an impedance of $0.1 + j1\ \Omega/\varphi$. The Y-connected source at the sending end of the line, has a positive phase sequence and an internal impedance of $0.02 + j0.2\ \Omega/\varphi$. Use the a-phase voltage at the load as the reference.

a) Draw three phase and single phase diagrams of the circuit.
b) Use circuit analysis techniques to compute line currents I_{aA}, I_{bB} and I_{cC}.
c) Calculate line voltages at the source terminals V_{ab}, V_{bc}, and V_{ca}.
d) Compute single phase and three phase active and reactive power delivered to the load.